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Gaussian Codes and the Scaled Nearest Neighbor Decoding in Fading Multi-Antenna Channels, Part 1: Information Theoretic Perspective

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ABSTRACT

We investigate the fading multi-antenna channel. The decoder is assumed to possess imperfect channel fading information. A modified nearest neighbor decoder is presented and an expression for the Generalized Mutual Information, the achievable rate, is obtained. We show that under certain conditions the achievable rate is equivalent to that of a fading multi-antenna Gaussian channel where fading is known to the receiver and is equal to the channel estimation, and where noise is due to both the channel noise and the channel estimation error. We show that for our communication scheme, the MMSE channel estimator is optimal in the sense that it achieves the highest value of Generalized Mutual Information, and hence the highest communication rate. We also investigate the outage capacity and show that when the channel estimation is not perfect there is a positive probability of outage, irrespective of the total transmit power.

Index Terms: Channel estimation, Fading channels, Gaussian code-books, Generalized Mutual Information, Multi-antenna channels, Outage capacity,
Scaled nearest neighbor decoder.